

Chapter 06

Switchable and Asymmetric Thermal Devices

From Static Metamaterials to Dynamically Reconfigurable Thermal Networks

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Pain Point

The static metamaterial trap. The transformation devices of Chapter 5—cloaks, concentrators, rotators—are fixed at fabrication. A wafer chuck optimized for one process recipe cannot adapt when the recipe changes. Semiconductor fabs run dozens of recipes on the same hardware, each requiring different thermal profiles. Without *switchable* thermal control, every recipe change demands hardware swaps, costing downtime and yield. Worse, heat flows in real systems are asymmetric: a die should shed heat easily to the heat sink, but parasitic back-flow from the board degrades performance. Static metamaterials treat heat flow as bidirectional; real applications need *thermal diodes* that impose directionality.

This chapter introduces two families of dynamic conduction devices—doublet metadevices for continuously tunable control and thermal diodes for asymmetric heat flow—and places both within the Mode-Path Matrix framework as network elements whose conductance can be switched, tuned, or made direction-dependent.

Learning Objectives

After completing this chapter, the reader will be able to:

- LO-06.1** Design a doublet metadevice that continuously tunes between cloaking and concentrating modes by specifying the anisotropic layer parameters and rotation angle.
- LO-06.2** Evaluate liquid metal thermal interface (LMTI) options for rotating and sliding metamaterial joints, including containment, corrosion, and reliability considerations.
- LO-06.3** Define the rectification ratio $\mathcal{R}$ and distinguish between shape-dependent, phase-change, and near-field radiative thermal diode mechanisms.
- LO-06.4** Design a cascaded thermal diode architecture to achieve enhanced rectification ratios for semiconductor applications.
- LO-06.5** Represent switchable and asymmetric devices as reconfigurable or nonlinear elements within the Mode-Path Matrix, including construction of the MPM for a system containing such devices.
- LO-06.6** Apply the six-step System Design Workflow (Chapter 4) to a thermal system incorporating switchable and directional elements, including critical-path identification and tool selection via Γ .

Abbreviations and Nomenclature

Acronyms

Abbreviation	Meaning
DR	Design Rule
EGaIn	Eutectic Gallium-Indium alloy
EMT	Effective Medium Theory
ESC	Electrostatic Chuck
FEA	Finite Element Analysis
GST	Ge ₂ Sb ₂ Te ₅ (phase-change alloy)
LMTI	Liquid Metal Thermal Interface
MIT	Metal-Insulator Transition
MPM	Mode-Path Matrix
PCM	Phase-Change Material
TIM	Thermal Interface Material
VO ₂	Vanadium Dioxide
WE	Worked Example

Principal Symbols

Symbol	Meaning	SI Unit
G	Thermal conductance	W K^{-1}
G_{cond}	Conduction conductance	W K^{-1}
G_{rad}	Linearized radiation conductance	W K^{-1}
G_{eff}	Effective (tunable) conductance	W K^{-1}
G_{fwd}	Forward-bias conductance	W K^{-1}
G_{rev}	Reverse-bias conductance	W K^{-1}
Γ	Mode dominance ratio ($G_{\text{rad}}/G_{\text{cond}}$)	—
$\kappa_{\parallel}$	Parallel thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$
$\kappa_{\perp}$	Perpendicular thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$
M	Conductance mismatch ratio	—
$\mathcal{R}$	Rectification ratio ($Q_{\text{fwd}}/Q_{\text{rev}}$)	—
R_{th}	Thermal resistance	K W^{-1}
R_{int}	Interface thermal resistance	$\text{m}^2 \text{K W}^{-1}$
R_{c}	Contact thermal resistance	$\text{m}^2 \text{K W}^{-1}$
θ	Relative rotation angle (doublet)	rad or deg
α	Fixed shell tilt angle	rad or deg
β	Bottleneck ratio	—
S	Thermal sensitivity (doublet)	rad^{-1}

Subscript/Superscript Conventions

Sub/Superscript	Meaning
fwd	Forward (high-conductance) direction
rev	Reverse (low-conductance) direction
eff	Effective value (doublet, averaged)

cond	Conduction mode
rad	Radiation mode
int	Interface

Numbering conventions in this chapter: Design Rules are labeled DR-06.*n*.¹ Worked Examples are WE-06.*n*. Learning Objectives are LO-06.*n*.

Chapter Roadmap and Deliverables

Engineering Deliverable: A switchable thermal device specification—doublet parameters, LMTI selection, or diode architecture—validated against the Mode-Path Matrix for a target semiconductor application.

This chapter extends the static conduction toolkit of Chapter 5 in two directions: *tunability* (Section 06.1, the doublet metadvice) and *asymmetry* (Section 06.3, thermal diodes and rectifiers). Both require the anisotropic materials designed in Chapter 5 and the network framework of Chapter 4.

The chapter workflow proceeds as follows:

1. Doublet operating principle and effective conductance derivation (§06.1).
2. Liquid metal thermal interfaces for rotating joints (§06.2).
3. Thermal diode mechanisms and rectification metrics (§06.3).
4. Device architectures and performance benchmarks (§06.4).
5. Network context: switchable and nonlinear elements in the MPM (§06.5).
6. Worked examples demonstrating MPM-integrated design (§06.6).

¹In earlier manuscript versions (v2.2), these rules were numbered DR-07.*n* and DR-08.*n* when this material occupied Chapters 7 and 8 separately. The mapping is: DR-06.1 = DR-07.1, DR-06.2 = DR-07.2, DR-06.3 = DR-07.3. No content has changed.

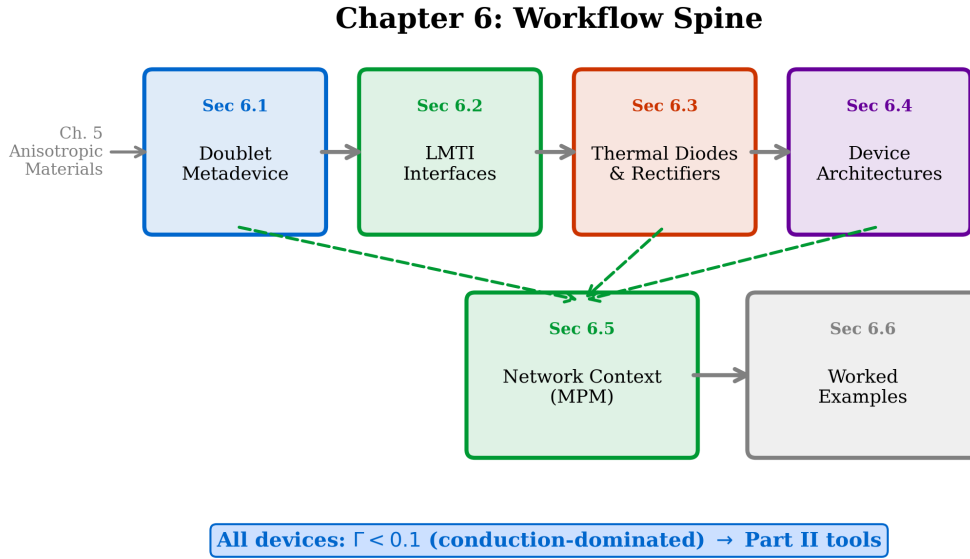


Figure 06.1: Chapter 6 workflow spine. The doublet metadvice (§06.1) and thermal diodes (§06.3) are both conduction-mode devices ($\Gamma < 0.1$) that modify the thermal network dynamically. The LMTI technology (§06.2) enables the mechanical interfaces required by the doublet. All devices are integrated into the Mode-Path Matrix framework (§06.5) as reconfigurable or nonlinear conductance elements.

06.1 The Doublet Metadevice

06.1.1 Why Static Metamaterials Are Not Enough

Chapter 5 demonstrated that transformation thermotics can redirect, concentrate, or shield heat flux with remarkable precision—provided the conductivity tensor is fabricated correctly. Every device in that chapter, however, shares a fundamental limitation: its thermal function is frozen into the material at the moment of manufacture. A five-layer Si/SiO₂ cloak will always cloak; it cannot be reconfigured into a concentrator when the process recipe changes. For laboratory demonstrations this is acceptable, but production semiconductor equipment must serve a far more demanding master: recipe diversity.

A modern 300 mm wafer fab typically runs 50–200 distinct process recipes on the same hardware platform. Each recipe specifies its own temperature profile: some require uniform heating to $\pm 0.5^\circ\text{C}$; others demand controlled radial gradients for film stress management; still others need rapid thermal cycling between different setpoints. The economic cost of recipe diversity falls squarely on the thermal hardware. With static thermal metamaterials, each recipe change requires either physical hardware swaps (hours of downtime) or compromise settings that sacrifice uniformity for versatility.

Beyond tunability, real thermal systems are fundamentally asymmetric. In a 3D chip stack, heat should flow freely from the die to the heat sink (forward direction), but parasitic heat flow from the board back into the die (reverse direction) degrades performance. Static metamaterials are inherently bidirectional: they modify heat flow equally in both directions. What the engineer needs is a thermal analog of the electrical diode—a device that permits heat flow in one direction while blocking it in the other.

This chapter addresses both needs. Section 06.1–06.2 introduces the *doublet metadvice*, which achieves continuously tunable thermal function by counter-rotating two anisotropic shells,

and the liquid metal thermal interface (LMTI) technology that makes the rotating joint production-viable. Section 06.3–06.4 develops *thermal diodes and rectifiers* based on shape-dependent, phase-change, and near-field mechanisms. Section 06.5 integrates both device families into the Mode-Path Matrix framework, showing how switchable and nonlinear conductance elements extend the network formalism of Chapter 4.

06.1.2 Operating Principle of the Doublet

Notation convention. In this chapter, α denotes the fixed tilt angle of each shell’s principal thermal axis relative to the radial direction (a fabrication parameter). θ denotes the relative rotation angle between inner and outer shells (the control variable). The shells rotate by $+\theta/2$ and $-\theta/2$ respectively, so the total relative displacement is θ .

Consider two concentric cylindrical shells, each fabricated from the binary layered composites of Chapter 5 (§05.4), with principal conductivities $\kappa_{\parallel}$ and $\kappa_{\perp}$ oriented at angles $+\alpha$ and $-\alpha$ relative to the radial direction. When the two shells are aligned ($\theta = 0$), their anisotropies reinforce, producing a concentrator. When counter-rotated to $\theta = \pi/2$, the anisotropies cancel, yielding an effectively isotropic (transparent) medium. At intermediate angles, the device interpolates continuously between these extremes, and for specific geometries can also approximate cloaking behavior.

The effective conductivity tensor of the doublet at rotation angle θ is:

$$\boldsymbol{\kappa}_{\text{eff}}(\theta) = \frac{1}{2} [\mathbf{R}(+\theta/2) \boldsymbol{\kappa}_0 \mathbf{R}^T(+\theta/2) + \mathbf{R}(-\theta/2) \boldsymbol{\kappa}_0 \mathbf{R}^T(-\theta/2)] \quad (06.1)$$

where $\boldsymbol{\kappa}_0 = \text{diag}(\kappa_{\parallel}, \kappa_{\perp})$ is the single-shell conductivity tensor and $\mathbf{R}(\phi)$ is the standard 2×2 rotation matrix.

Expanding Eq. (06.1) yields the diagonal components:

$$\kappa_{rr}^{\text{eff}}(\theta) = \kappa_{\parallel} \cos^2(\theta/2) + \kappa_{\perp} \sin^2(\theta/2) \quad (06.2)$$

$$\kappa_{\theta\theta}^{\text{eff}}(\theta) = \kappa_{\parallel} \sin^2(\theta/2) + \kappa_{\perp} \cos^2(\theta/2) \quad (06.3)$$

The effective anisotropy ratio is then:

$$\frac{\kappa_{\theta\theta}^{\text{eff}}}{\kappa_{rr}^{\text{eff}}} = \frac{\kappa_{\parallel} \sin^2(\theta/2) + \kappa_{\perp} \cos^2(\theta/2)}{\kappa_{\parallel} \cos^2(\theta/2) + \kappa_{\perp} \sin^2(\theta/2)} \quad (06.4)$$

Three limiting cases emerge:

- $\theta = 0$ (aligned): $\kappa_{rr}^{\text{eff}} = \kappa_{\parallel}$, $\kappa_{\theta\theta}^{\text{eff}} = \kappa_{\perp}$. Maximum radial conductivity $\rightarrow$ **concentrator** mode.
- $\theta = \pi$ (counter-aligned): $\kappa_{rr}^{\text{eff}} = \kappa_{\perp}$, $\kappa_{\theta\theta}^{\text{eff}} = \kappa_{\parallel}$. Maximum tangential conductivity $\rightarrow$ **cloak** mode.
- $\theta = \pi/2$ (orthogonal): $\kappa_{rr}^{\text{eff}} = \kappa_{\theta\theta}^{\text{eff}} = (\kappa_{\parallel} + \kappa_{\perp})/2$. Isotropic $\rightarrow$ **transparent** mode.

Figure 06.2 illustrates these three modes schematically, showing the heat flux pattern in each configuration.

Figure 06.2: Doublet Metadevice Operating Modes

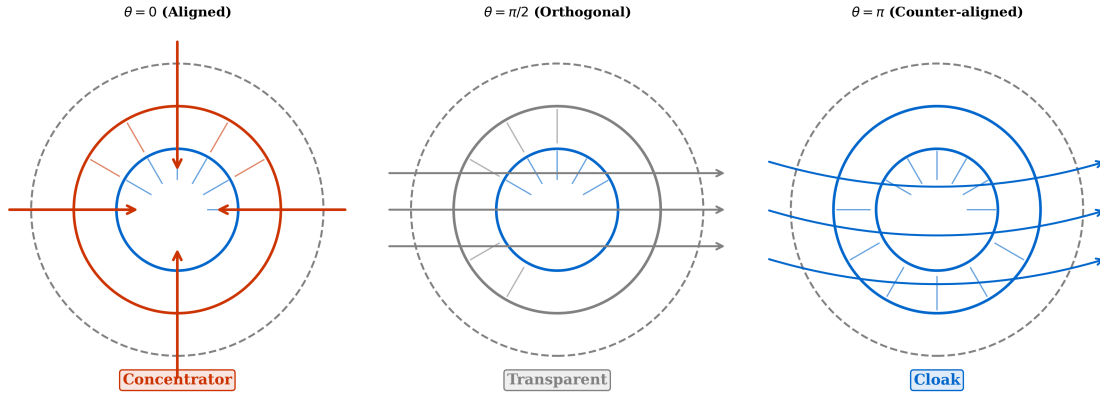


Figure 06.2: Doublet metadevice operating modes. **Left:** $\theta = 0$ (aligned)—anisotropies reinforce, producing radial flux concentration. **Center:** $\theta = \pi/2$ (orthogonal)—anisotropies cancel, yielding isotropic (transparent) response. **Right:** $\theta = \pi$ (counter-aligned)—tangential conductivity dominates, diverting flux around the inner region (cloaking). The inner shell (blue) is fixed; the outer shell (colored) rotates. Arrows indicate heat flux direction and magnitude.

Conduction Mode (Diffusionics)

The doublet is a conduction-mode device. It operates on the conductivity tensor κ , not on emissivity or radiative properties. In the Mode-Path Matrix, it modifies G_{cond} on the path it occupies. The doublet is the correct tool only when $\Gamma < 0.1$ on the critical path (DR-05.6). At high temperatures where $\Gamma > 0.1$, the doublet's effect on total heat transfer diminishes because the radiation channel bypasses the conduction modification.

06.1.3 Effective Conductance as a Function of Rotation Angle

Units Reminder

κ	$[\text{W m}^{-1} \text{K}^{-1}]$	Thermal conductivity (material, intensive)
G	$[\text{W K}^{-1}]$	Thermal conductance (path, extensive: $G = \kappa A/L$)
R_{th}	$[\text{K W}^{-1}]$	Thermal resistance ($R_{\text{th}} = 1/G$)
R_{int}	$[\text{m}^2 \text{K W}^{-1}]$	Interface thermal resistance ($G = A/R_{\text{int}}$)

Doubling cross-section area doubles G but does not change κ . Always verify which quantity is being compared.

For a doublet shell with inner radius R_1 , outer radius R_2 , and length L (in a cylindrical geometry), the effective radial thermal conductance from the inner to outer surface is:

$$G_{\text{eff}}(\theta) = \frac{2\pi L \sqrt{\kappa_{rr}^{\text{eff}}(\theta) \kappa_{\theta\theta}^{\text{eff}}(\theta)}}{\ln(R_2/R_1)} \quad (06.5)$$

Substituting Eqs. (06.2)–(06.3):

$$G_{\text{eff}}(\theta) = \frac{2\pi L}{\ln(R_2/R_1)} \sqrt{[\kappa_{\parallel} \cos^2 \frac{\theta}{2} + \kappa_{\perp} \sin^2 \frac{\theta}{2}] [\kappa_{\parallel} \sin^2 \frac{\theta}{2} + \kappa_{\perp} \cos^2 \frac{\theta}{2}]} \quad (06.6)$$

The tuning range of the doublet is characterized by:

$$\frac{G_{\text{max}}}{G_{\text{min}}} = \frac{G_{\text{eff}}(0)}{G_{\text{eff}}(\pi)} = \frac{\kappa_{\parallel}}{\kappa_{\perp}} = \kappa_{\parallel}/\kappa_{\perp} \quad (06.7)$$

This is a remarkable result: the conductance tuning range equals the anisotropy ratio of the constituent shells.

Design Rule:

Anisotropy Definitions — Resolving κ_A/κ_B vs. $\kappa_{\parallel}/\kappa_{\perp}$ vs. $G_{\max}/G_{\min}$ Four distinct anisotropy-related quantities appear in this book. They are *not* interchangeable:

(A) **Constituent conductivity contrast** κ_A/κ_B : the ratio of bulk thermal conductivities of the two materials forming the binary composite. This is a *material property*, independent of geometry or fill fraction.

Examples: Si/SiO₂: $\kappa_A/\kappa_B = 130/1.4 \approx 93$. Cu/Kapton: $\kappa_A/\kappa_B = 390/0.12 \approx 3250$.

(B) **EMT anisotropy at transformation fill fraction** $(\kappa_{\parallel}/\kappa_{\perp})_{f_{\text{opt}}}$: the anisotropy ratio at the fill fraction $f_{\text{opt}} = 1/(1 + \sqrt{\kappa_A/\kappa_B})$ from Eq. (05.15). This fill fraction is optimized for transformation thermotics device design (matching specific $\kappa_{\parallel}$ and $\kappa_{\perp}$ targets to the Jacobian prescription), *not* for maximizing the anisotropy ratio itself.

Examples at f_{opt} : Si/SiO₂ ($f_{\text{opt}} = 0.094$): $(\kappa_{\parallel}/\kappa_{\perp})_{f_{\text{opt}}} \approx 8.7$. Cu/Kapton ($f_{\text{opt}} = 0.017$): $(\kappa_{\parallel}/\kappa_{\perp})_{f_{\text{opt}}} \approx 56$.

(B') **Maximum EMT anisotropy** $(\kappa_{\parallel}/\kappa_{\perp})_{\max}$: the *true* maximum of $\kappa_{\parallel}/\kappa_{\perp}$ over all fill fractions, which occurs at $f = 0.5$ for *every* binary pair. From Eq. (05.14):

$$\left(\frac{\kappa_{\parallel}}{\kappa_{\perp}}\right)_{\max} = 1 + \frac{1}{4} \left(\frac{\kappa_A}{\kappa_B} + \frac{\kappa_B}{\kappa_A} - 2 \right) \quad \text{at } f = 0.5 \quad (06.8)$$

Proof: $\kappa_{\parallel}/\kappa_{\perp} = 1 + f(1-f)(r+1/r-2)$ where $r = \kappa_A/\kappa_B$. Since $r+1/r-2 > 0$ for $r \neq 1$, and $f(1-f)$ is maximized at $f = 0.5$, the maximum of $\kappa_{\parallel}/\kappa_{\perp}$ occurs at $f = 0.5$. QED.

Examples at $f = 0.5$: Si/SiO₂: $(\kappa_{\parallel}/\kappa_{\perp})_{\max} \approx 24$. Cu/Kapton: $(\kappa_{\parallel}/\kappa_{\perp})_{\max} \approx 813$. Ge/As₂Se₃: $(\kappa_{\parallel}/\kappa_{\perp})_{\max} \approx 76$.

(C) **Device tuning range** $G_{\max}/G_{\min}$: the ratio of maximum to minimum thermal conductance achievable by the doublet through rotation. From Eq. (06.7), this equals the EMT anisotropy of the constituent shells *at the chosen fill fraction* f .

For doublet applications where maximum tuning range is the primary goal, use $f = 0.5$ (category B'). For transformation devices where specific $\kappa_{\parallel}$ and $\kappa_{\perp}$ values must match a Jacobian prescription, use f_{opt} (category B) and accept the reduced tuning range.

Convention: The unqualified symbol $\kappa_{\parallel}/\kappa_{\perp}$ in this book refers to the EMT anisotropy at the *design fill fraction* (whichever f is chosen). The maximum value is always denoted $(\kappa_{\parallel}/\kappa_{\perp})_{\max}$ with the subscript “max.” The raw contrast κ_A/κ_B is always written explicitly.

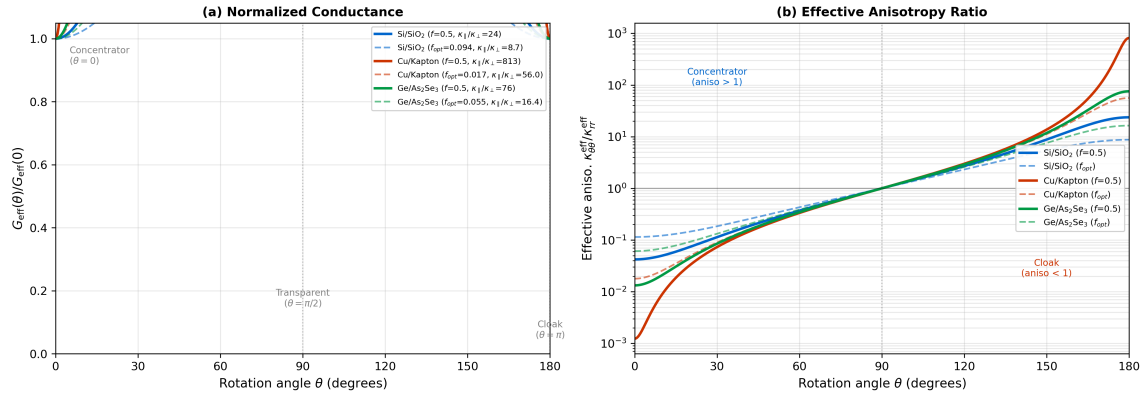
Figure 06.3: Effective Radial Conductance vs. Rotation Angle (Corrected EMT Values)

Figure 06.3: Effective radial conductance versus rotation angle for three material pairs. **Solid lines:** $f = 0.5$ (maximum-anisotropy fill fraction, category B'). **Dashed lines:** f_{opt} from Eq. (05.15) (transformation-device fill fraction, category B). **(a)** $G_{\text{eff}}(\theta)/G_{\text{eff}}(0)$: Si/SiO₂ ($\kappa_{\parallel}/\kappa_{\perp} = 24$ at $f = 0.5$; 8.7 at $f_{\text{opt}} = 0.094$), Cu/Kapton ($\kappa_{\parallel}/\kappa_{\perp} = 813$ at $f = 0.5$; 56 at $f_{\text{opt}} = 0.017$), Ge/As₂Se₃ ($\kappa_{\parallel}/\kappa_{\perp} = 76$ at $f = 0.5$; 16 at $f_{\text{opt}} = 0.055$). The smooth $\cos^2(\theta/2)$ dependence enables precise analog tuning. **(b)** Effective anisotropy ratio on log scale, showing inversion from concentrator (> 1) to cloak (< 1) as θ passes through $\pi/2$. See Anisotropy Definitions box for the distinction between κ_A/κ_B , $(\kappa_{\parallel}/\kappa_{\perp})_{f_{\text{opt}}}$, and $(\kappa_{\parallel}/\kappa_{\perp})_{\text{max}}$.

06.1.4 Mechanical Design Considerations

The doublet requires a rotatable interface between the inner and outer shells. This rotating joint introduces a contact thermal resistance that can severely degrade device performance. Three interface strategies are available, summarized in Table 06.4.

Table 06.4: Interface fit types for doublet metadvice.

Fit Type	R_{th} Penalty	Torque	Application
Interference	$< 5\%$	10–50×	Static or rare switching
Clearance (air)	30–100%	1×	Frequent switching, low precision
Clearance + LMTI	5–15%	2–3×	Production recommended

The clearance-with-LMTI configuration is recommended for production applications because it balances low thermal resistance penalty (5–15%) against reasonable actuation torque and unlimited rotation cycles. The LMTI technology is treated in detail in Section 06.2.

Design Rule:

DR-06.1 — Doublet Interface Selection For doublet metadvice requiring > 10 switching cycles over the device lifetime, use a clearance fit with LMTI. Verify that the interface thermal resistance penalty $\Delta R_{\text{th}}/R_{\text{th}} < 0.15$ (i.e., $< 15\%$ degradation of the doublet conductance tuning range). For static or rarely switched applications (< 10 cycles), interference fits provide superior thermal performance.

06.1.5 Tolerancing the Doublet: The Thermal Zoom Analogy

The doublet metadvice is functionally a *thermal zoom system*: the rotation angle θ is the zoom parameter and $G_{\text{eff}}(\theta)$ is the thermal “magnification.” This analogy, familiar to optical engineers (Book 1, Ch. 9), provides immediate insight into doublet tolerancing.

Define the *thermal sensitivity* at operating point θ_0 :

$$S(\theta_0) = \left| \frac{1}{G_{\text{eff}}} \frac{dG_{\text{eff}}}{d\theta} \right|_{\theta_0} \quad (06.9)$$

This is the fractional conductance change per radian of rotation error—the thermal analog of the zoom sensitivity dF/dz in lens design (Code V ZSE operand).

From Eq. (06.6), the sensitivity evaluates to:

$$S(\theta) = \frac{(\kappa_{\parallel} - \kappa_{\perp})^2 |\sin \theta|}{4 [\kappa_{\parallel} \cos^2(\theta/2) + \kappa_{\perp} \sin^2(\theta/2)] [\kappa_{\parallel} \sin^2(\theta/2) + \kappa_{\perp} \cos^2(\theta/2)]} \quad (06.10)$$

Figure 06.4 plots $S(\theta)$ and the corresponding thermal depth of focus for three representative material pairs.

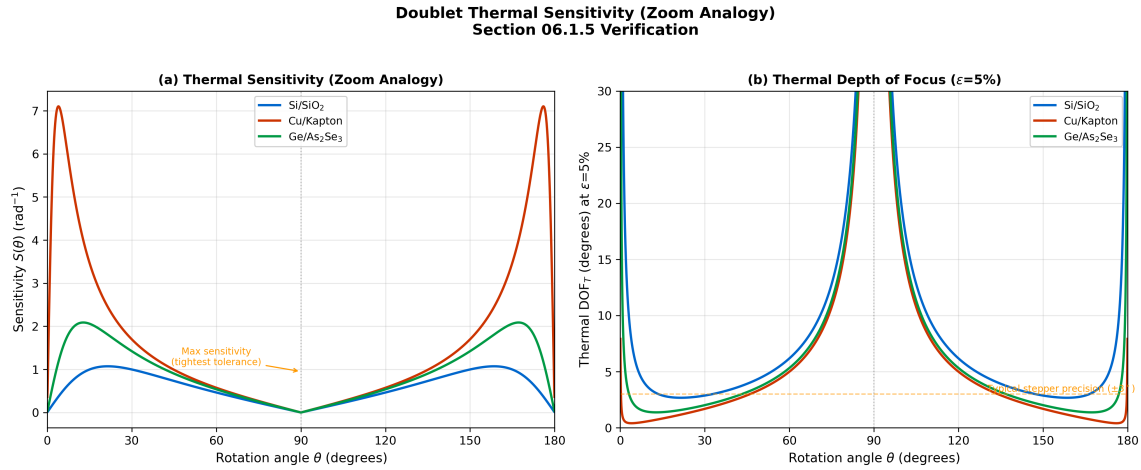


Figure 06.4: Doublet thermal sensitivity (zoom analogy), Section 06.1.5 verification. **(a)** Thermal sensitivity $S(\theta)$ from Eq. (06.10) for Si/SiO₂ ($\kappa_{\parallel}/\kappa_{\perp} = 24$), Cu/Kapton ($\kappa_{\parallel}/\kappa_{\perp} = 813$), and Ge/As₂Se₃ ($\kappa_{\parallel}/\kappa_{\perp} = 76$), all at $f = 0.5$. Maximum sensitivity occurs near $\theta = \pi/2$ (transparent mode), where control precision is most critical—the thermal analog of mid-zoom in a zoom lens. $S(0) = S(\pi) = 0$ at the concentrator and cloak endpoints. **(b)** Thermal depth of focus $\text{DOF}_T = \epsilon/S(\theta)$ at $\epsilon = 5\%$ tolerance. The dashed orange line marks a typical rotary stepper precision ($\pm 3^\circ$). Cu/Kapton, with the highest anisotropy, has the narrowest DOF_T and requires the finest angular control.

Key properties:

- $S(0) = S(\pi) = 0$: At the concentrator and cloak endpoints, the system is at a local extremum and first-order insensitive to rotation error.
- $S(\pi/2) = S_{\text{max}}$: The transparent mode has maximum sensitivity—control precision requirements are most stringent here.
- For high-anisotropy materials ($\kappa_{\parallel}/\kappa_{\perp} \gg 1$): $S_{\text{max}} \approx (\kappa_{\parallel}/\kappa_{\perp} - 1)/(\kappa_{\parallel}/\kappa_{\perp} + 1) \approx 1 \text{ rad}^{-1}$. A 1° (0.017 rad) error at $\theta = \pi/2$ causes $\sim 1.7\%$ conductance error.

Define the *thermal depth of focus* (DOF_T) as the angular range over which G_{eff} stays within a tolerance ϵ of the target value:

$$\text{DOF}_T(\theta_0, \epsilon) = \frac{\epsilon}{S(\theta_0)} \quad (06.11)$$

Table 06.5: Thermal depth of focus for a Si/SiO₂ doublet ($\kappa_{\parallel}/\kappa_{\perp} = 24$ at $f = 0.5$) with $\epsilon = 5\%$ tolerance.

Operating Point	θ	S (rad ⁻¹)	DOF_T (deg)
Concentrator	0	0	∞ (at extremum)
Quarter-turn	$\pi/4$	0.72	4.0°
Transparent	$\pi/2$	0.92	3.1°
Three-quarter	$3\pi/4$	0.72	4.0°
Cloak	π	0	∞ (at extremum)

Engineering implication: Near $\theta = \pi/2$ (transparent mode), the actuation system must achieve angular precision better than $\pm 3^\circ$ to maintain 5% conductance tolerance. At the concentrator and cloak endpoints, tolerance is relaxed. This mirrors optical zoom systems where the tightest mechanical tolerances occur at intermediate zoom positions.

Trilogy Connection:

Eikonal Bridge Ch. 9 (Zoom Lens Tolerancing) In a zoom lens, the compensator sensitivity $\partial \text{MTF} / \partial z$ peaks at mid-zoom positions where element spacings change most rapidly. Optical designers use “worst-case zoom” analysis to set mechanical tolerances. The thermal doublet’s $S(\theta)$ curve plays an identical role: tolerance budgets should be set at $\theta = \pi/2$ (the “worst-case zoom” of the thermal system), not at the endpoints.

06.2 Liquid Metal Thermal Interfaces

06.2.1 Material Options

Liquid metal thermal interfaces (LMTIs) fill the clearance gap in rotating or sliding metamaterial joints with a high-conductivity liquid, dramatically reducing the contact resistance compared to an air gap.

Two alloy families dominate LMTI applications:

Table 06.6: Properties of LMTI candidate alloys.

Alloy	T_m (°C)	κ (W/m·K)	Viscosity (mPa·s)	Toxicity
Galinstan (Ga-In-Sn)	−19	16.5	2.4	Low
EGaIn (Ga-In)	15.5	26.4	1.99	Low
Ga pure	29.8	40.6	1.81	Low
Hg (reference)	−38.8	8.3	1.55	High—banned

Galinstan is the preferred choice for most applications due to its low melting point (liquid at room temperature), reasonable thermal conductivity (16.5 W/m·K, compared to 0.026 W/m·K for air), and low toxicity. EGaIn offers higher conductivity but solidifies at 15.5 °C, limiting its use in cold environments.

The thermal conductance of an LMTI gap of thickness d over area A is:

$$G_{\text{LMTI}} = \frac{\kappa_{\text{LMTI}} \cdot A}{d} \quad (06.12)$$

For a typical doublet clearance ($d = 50 \mu\text{m}$, $A = 10^{-3} \text{ m}^2$, Galinstan):

$$G_{\text{LMTI}} = \frac{16.5 \times 10^{-3}}{50 \times 10^{-6}} = 330 \text{ W/K} \quad (06.13)$$

Compared to an air gap of the same dimensions ($G_{\text{air}} = 0.026 \times 10^{-3}/(50 \times 10^{-6}) = 0.52 \text{ W/K}$), the LMTI provides a $630\times$ improvement.

06.2.2 Containment and Corrosion

Gallium-based alloys aggressively wet most metals, creating intermetallic compounds that compromise structural integrity. Figure 06.5 shows the cross-section of a production LMTI joint and the associated thermal resistance stack. The key engineering challenges are:

1. **Gallium embrittlement:** Ga attacks Al, Cu, and Zn grain boundaries. *Mitigation:* Use Ni or W barrier coatings ($> 2 \mu\text{m}$ thickness) on all surfaces contacting the LMTI.
2. **Oxide skin formation:** Ga_2O_3 forms a rigid skin on exposed surfaces, increasing apparent viscosity and impeding wetting. *Mitigation:* Operate in inert atmosphere (N_2 or Ar) or apply light HCl vapor treatment during assembly.
3. **Containment:** LMTIs must be sealed against leakage, especially in rotating joints. *Mitigation:* O-ring seals with fluoroelastomer (FKM) or PTFE materials; maintain gap uniformity to prevent hydraulic pumping during rotation.

Figure 06.4: Liquid Metal Thermal Interface (LMTI) Design

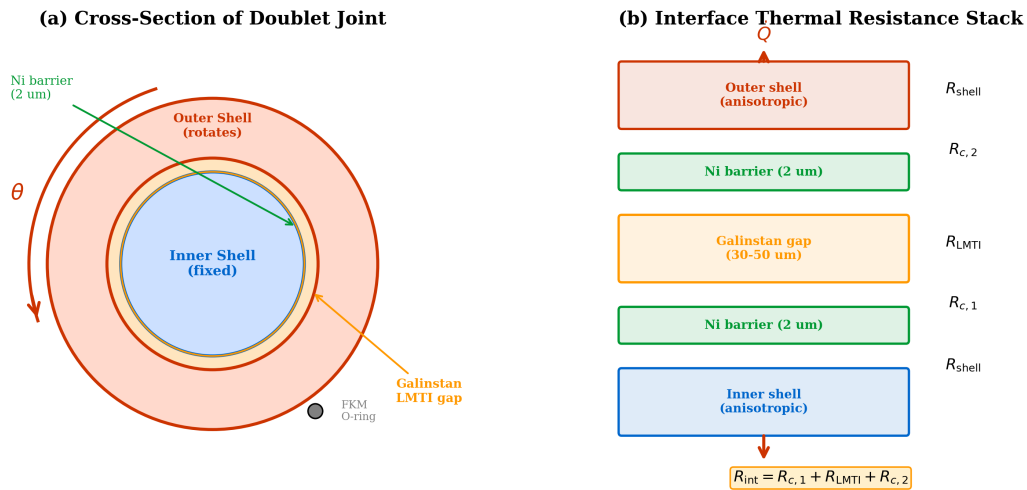


Figure 06.5: Liquid metal thermal interface (LMTI) design. (a) Cross-section of a doublet rotating joint showing inner shell (fixed, blue), outer shell (rotates, red), Galinstan LMTI gap (orange), Ni barrier coatings, and FKM O-ring seals. (b) Thermal resistance stack: $R_{\text{int}} = R_{c,1} + R_{\text{LMTI}} + R_{c,2}$ (Eq. 06.14). The LMTI bulk resistance dominates for gaps $d > 10 \mu\text{m}$.

Fabrication Challenge

Long-term LMTI reliability. Published data on Galinstan LMTIs show $< 5\%$ conductance degradation after 10,000 rotation cycles at room temperature. However, data above 200°C is limited. For semiconductor applications operating at elevated temperatures, accelerated aging tests (JEDEC JESD22-A104-style thermal cycling) are essential before production deployment. The failure mode is typically Ni barrier breakdown followed by intermetallic growth, increasing R_{int} by $2\text{--}5\times$ over 6–12 months (see Chapter 12, Case Study CS-1).

LMTI Qualification Test Plan

Objective: Demonstrate $G_{\text{LMTI}} > 10 \times G_{\text{doublet,min}}$ (DR-06.2) after accelerated aging.

Test	Specification
Temperature range	Cycle between $T_{\text{min}} = 25^\circ\text{C}$ and $T_{\text{max}} = \text{operating temp.} + 50^\circ\text{C}$ (e.g., 400°C for wafer chucks). Per JEDEC JESD22-A104 profile.
Cycle count	Min 10,000 cycles (standard); 100,000 cycles (high-throughput: > 50 recipe changes/day over 5-year service life).
Inspection metrics	(a) G_{LMTI} at 0, 100, 1k, 5k, 10k cycles via transient measurement. (b) Cross-section SEM of Ni barrier interface. (c) EDS mapping for Ga diffusion depth.
Pass/fail	(a) $\Delta G_{\text{LMTI}}/G_{\text{LMTI},0} < 15\%$. (b) No Ga penetration through Ni barrier (intermetallic $< t_{\text{Ni}}/2$). (c) No visible seal leakage.

06.2.3 Contact Resistance Engineering

The total thermal resistance across a doublet rotating interface consists of three series contributions:

$$R_{\text{interface}} = R_{c,1} + R_{\text{LMTI}} + R_{c,2} \quad (06.14)$$

where $R_{c,1}$ and $R_{c,2}$ are the solid-liquid contact resistances at each wetted surface, and $R_{\text{LMTI}} = d/(\kappa_{\text{LMTI}}A)$ is the bulk liquid resistance.

The contact resistance depends strongly on surface finish. As an *order-of-magnitude scaling estimate* (not a derived law):

$$R_c \approx \frac{R_a}{\kappa_{\text{LMTI}}} \quad [\text{scaling estimate}] \quad (06.15)$$

where R_a is the arithmetic mean roughness. This estimate assumes: (i) the liquid metal fully wets surface asperities (contact angle $< 90^\circ$), (ii) the roughness profile is approximately Gaussian, and (iii) asperity height $\ll$ LMTI gap thickness d . For $R_a < 0.8 \mu\text{m}$ with Galinstan ($\kappa_{\text{LMTI}} = 16.5 \text{ W/m}\cdot\text{K}$): $R_c \approx 0.8 \times 10^{-6}/16.5 \approx 5 \times 10^{-8} \text{ m}^2\text{K/W}$, negligible compared to the bulk LMTI resistance for gaps $d > 10 \mu\text{m}$.

Design Rule:

DR-06.2 — LMTI Surface Preparation For LMTI-filled rotating joints: (a) apply Ni barrier coating $\geq 2 \mu\text{m}$ on all Ga-contacting surfaces, (b) maintain surface finish $R_a < 0.8 \mu\text{m}$, (c) assemble in inert atmosphere or apply oxide removal treatment, (d) use FKM or PTFE seals. Verify $G_{\text{LMTI}} > 10 \times G_{\text{doublet,min}}$ to ensure the interface is not the bottleneck.

06.3 Thermal Diodes and Rectifiers

06.3.1 Asymmetric Heat Flow: Concepts and Metrics

A thermal diode is a device that conducts heat preferentially in one direction. The key performance metric is the *rectification ratio*:

$$\mathcal{R} = \frac{Q_{\text{fwd}}}{Q_{\text{rev}}} \quad (06.16)$$

where Q_{fwd} is the heat flow when the temperature gradient is in the “forward” (high-conductance) direction, and Q_{rev} is the heat flow under the same temperature difference but reversed gradient. An ideal diode has $\mathcal{R} \rightarrow \infty$; a symmetric conductor has $\mathcal{R} = 1$.

Equivalently, one can define forward and reverse conductances:

$$G_{\text{fwd}} = \frac{Q_{\text{fwd}}}{\Delta T}, \quad G_{\text{rev}} = \frac{Q_{\text{rev}}}{\Delta T}, \quad \mathcal{R} = \frac{G_{\text{fwd}}}{G_{\text{rev}}} \quad (06.17)$$

Trilogy Connection:

Eikonal Bridge Ch. 8 (Optical Isolator Analogy) A thermal diode is the heat-flow analog of an *optical isolator* (Faraday rotator + polarizer). Both break time-reversal symmetry: the optical isolator uses magneto-optic nonreciprocity; the thermal diode uses temperature-dependent conductivity (a material nonlinearity). Optical isolators routinely achieve $> 1000\times$ suppression ($> 30 \text{ dB}$), while thermal diodes remain orders of magnitude behind ($\mathcal{R} = 1.5\text{--}50$)—a gap that cascading (Eq. 06.20) partially closes.

Figure 06.6 illustrates the rectification principle, showing the asymmetry between forward-bias (phase-change material in the high-conductivity state) and reverse-bias (low-conductivity state) operation, alongside the electrical diode analogy.

Figure 06.5: Thermal Diode Rectification Concept

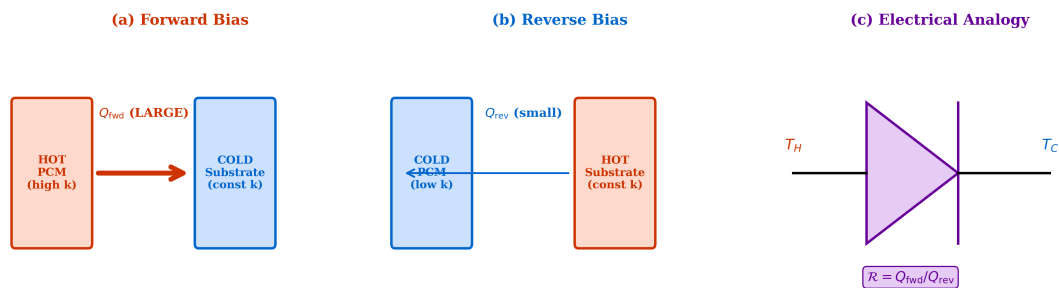


Figure 06.6: Thermal diode rectification concept. **(a)** Forward bias: the phase-change material (PCM) is heated above its transition temperature, entering the high- κ state; heat flows freely (Q_{fwd} large). **(b)** Reverse bias: the PCM remains below transition; heat flow is restricted (Q_{rev} small). **(c)** Electrical diode analogy: the I-V characteristic maps to the Q - ΔT characteristic.

06.3.2 Shape-Dependent Thermal Rectification

The simplest thermal diode exploits geometry: two dissimilar materials joined in series, where both have temperature-dependent thermal conductivities with opposite slopes ($d\kappa/dT > 0$ for material A, $d\kappa/dT < 0$ for material B). When the hot side contacts the material whose κ increases with temperature, forward conductance is high; when reversed, both materials are in their low- κ states.

Shape-dependent diodes achieve $\mathcal{R} \approx 1.1$ – 1.5 in practice, limited by the modest $d\kappa/dT$ of common solids. Their primary advantage is simplicity and high forward conductance.

06.3.3 Phase-Change Material Thermal Diodes

06.3.3.1 Vanadium Dioxide (VO₂)

VO₂ undergoes a metal-insulator transition (MIT) at $T_{\text{MIT}} \approx 68^\circ\text{C}$, with thermal conductivity changing from $\kappa_{\text{ins}} \approx 3.5$ W/m·K (insulating phase) to $\kappa_{\text{met}} \approx 6$ W/m·K (metallic phase)—a factor of ~ 1.7 .

A thermal diode is constructed by placing VO₂ in series with a material of constant κ :

- **Forward bias:** Hot side on VO₂ end. VO₂ is above MIT $\rightarrow$ high κ . Total conductance is high.
- **Reverse bias:** Hot side on constant- κ end. VO₂ remains below MIT $\rightarrow$ low κ . Total conductance is low.

In practice, VO₂-based diodes achieve $\mathcal{R} \approx 1.5$ – 2.0 for thin-film configurations.

06.3.3.2 Ge₂Sb₂Te₅ (GST)

GST phase-change alloys, widely used in optical data storage, offer a much larger conductivity contrast: $\kappa_{\text{amorphous}} \approx 0.2$ W/m·K versus $\kappa_{\text{crystalline}} \approx 1.5$ – 2.0 W/m·K, giving a contrast ratio of ~ 8 – 10 .

The transition temperature (~ 150 – 170°C for crystallization) is tunable through composition, and the phase change can be triggered electrically (Joule heating) for active control. GST-based thermal diodes have demonstrated $\mathcal{R} \approx 3$ – 5 in laboratory settings.

Table 06.7: Comparison of phase-change thermal diode materials.

Material	T_{switch} ($^\circ\text{C}$)	κ_{low} (W/m·K)	κ_{high} (W/m·K)	$\mathcal{R}$ (demo)
VO ₂	68	3.5	6.0	1.5–2.0
GST	150–170	0.2	1.5–2.0	3–5
Nitinol (SMA)	70–100	8.6	18	1.5–2.1

Figure 06.6: Phase-Change Material Thermal Conductivity

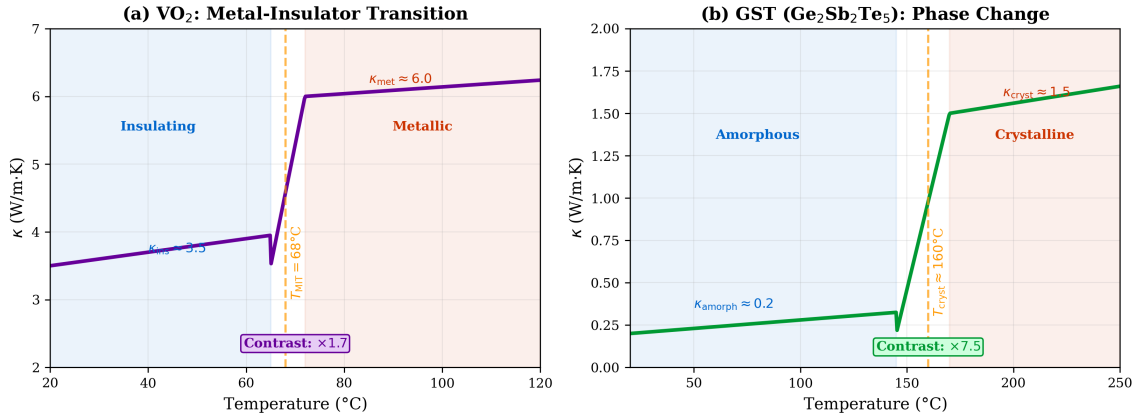


Figure 06.7: Phase-change thermal conductivity profiles. **(a)** VO₂: metal-insulator transition at $T_{MIT} \approx 68^\circ\text{C}$ produces a $1.7\times$ conductivity contrast. **(b)** GST (Ge₂Sb₂Te₅): amorphous-to-crystalline transition at $\sim 160^\circ\text{C}$ produces a $7.5\times$ contrast—the basis for higher rectification ratios in cascaded diode architectures.

06.3.4 Near-Field Radiative Thermal Diodes

When two surfaces are separated by a sub-wavelength gap ($d < \lambda_{th} \approx 10 \mu\text{m}$ at 300 K), near-field radiative transfer (Chapter 3) can exceed the blackbody limit by orders of magnitude. If one surface is a phase-change material, the near-field exchange becomes strongly asymmetric:

$$\mathcal{R}_{NF} = \frac{q_{NF}(T_1 > T_{MIT}, T_2 < T_{MIT})}{q_{NF}(T_2 > T_{MIT}, T_1 < T_{MIT})} \quad (06.18)$$

Near-field VO₂ diodes have demonstrated $\mathcal{R} > 10$ at gap distances of 100–500 nm. However, maintaining sub-micron gaps over practical areas remains an extreme fabrication challenge (Chapter 13), and these devices are currently research-grade rather than production-ready.

Multiphysics Integration

Near-field diodes cross mode boundaries. Unlike the conduction-mode devices in Sections 06.3.2–06.3.3, near-field diodes operate via radiation ($\Gamma \gg 1$ across the gap) but with conduction-like gap dependence ($q \propto 1/d^2$ rather than $q \propto T^4$). In the Mode-Path Matrix, the gap path has $\Gamma > 10$, but the device as a whole may sit on a conduction-dominated critical path if the gap conductance exceeds the adjacent solid conductances. The Γ classification still applies to each *path segment* independently.

Table 06.8: Near-field thermal diode feasibility envelope for production assessment.

Parameter	Current Range	Production Constraint
Gap distance d	50–500 nm	Requires active gap control or precision spacers (SiO ₂ pillars, MEMS actuators)
Planarity tolerance	± 10 nm	CMP-grade surface finish; incompatible with rough or warped substrates
Device area	< 1 mm ²	Gap uniformity degrades above $\sim$ mm ² ; wafer-scale not demonstrated
Contamination risk	High	Any particle $> d$ causes contact and shorts the diode; cleanroom Class 100 or better
Spacer strategy	Oxide pillars	Pillar arrays reduce $\mathcal{R}$ by providing conduction shunts; trade-off between stability and rectification
Γ regime	> 1 (gap path)	Gap is radiation-dominated; verify surrounding paths do not shunt the diode conductively (check MPM)
Production grade	C (Research)	Not production-ready. Laboratory demonstrations only.

Radiation Mode (Thermal Photonics)

Near-field diode: Γ classification. Unlike bulk conduction diodes, a near-field radiative diode operates in a *radiation-dominated* gap. If the surrounding structure provides a parallel conduction path with $G_{\text{cond}} \gg G_{\text{rad}}^{\text{NF}}$, the diode is shunted and $\mathcal{R}_{\text{system}} \rightarrow 1$. Before specifying a near-field diode, construct the local MPM: the diode edge must have $\Gamma > 1$ *and* must lie on or near the critical path. This is the inverse of the doublet requirement ($\Gamma < 0.1$)—a near-field diode is a Part III (thermal photonics) device, not a Part II (diffusionics) device.

06.4 Device Architectures

06.4.1 1D Flat-Plate Configurations

For wafer chuck applications, the cylindrical doublet geometry of Section 06.1 must be translated to a planar (Cartesian) architecture. The planar doublet consists of two stacked anisotropic plates with in-plane conductivities $\kappa_{\parallel}$ and $\kappa_{\perp}$, where relative sliding (rather than rotation) tunes the effective through-thickness conductance.

The effective through-plane conductance of two stacked anisotropic plates, each of thickness t , with the high- κ axis of the upper plate rotated by angle θ relative to the lower:

$$G_{\perp}^{\text{eff}}(\theta) = \frac{A}{2t} \frac{2\kappa_{\perp}\kappa_{\parallel}}{\kappa_{\parallel}\sin^2\theta + \kappa_{\perp}\cos^2\theta + \kappa_{\parallel}\cos^2\theta + \kappa_{\perp}\sin^2\theta} = \frac{A}{2t} \frac{2\kappa_{\perp}\kappa_{\parallel}}{\kappa_{\parallel} + \kappa_{\perp}} \quad (06.19)$$

Note that the through-plane conductance is *independent of θ* for homogeneous stacked plates with no lateral heat spreading. Tuning requires either: (a) creating lateral gradients that couple in-plane anisotropy to through-plane flow, or (b) using a checkerboard pattern of meta-elements (“doublet array”) where each element’s rotation is individually controlled.

Design Rule:

DR-06.3 — Planar Doublet Array Design For planar (Cartesian) doublet arrays: use a meta-element grid of minimum 4×4 elements; set pitch $\leq$ the target spatial resolution of the thermal profile. Verify that edge meta-elements have boundary compensation (guard ring or graded angle offset) to suppress thermal “aberrations” at the array periphery.

06.4.2 Cascaded Thermal Diode Stages

A single thermal diode stage provides limited rectification ($\mathcal{R} = 1.5\text{--}5$ for passive devices). Cascading N stages in series amplifies the rectification ratio:

$$\mathcal{R}^{(N)} = \prod_{i=1}^N \mathcal{R}_i \quad (06.20)$$

For identical stages: $\mathcal{R}^{(N)} = \mathcal{R}_1^N$. With GST stages ($\mathcal{R}_1 = 3$):

$$N = 2 : \mathcal{R} = 9, \quad N = 3 : \mathcal{R} = 27, \quad N = 4 : \mathcal{R} = 81 \quad (06.21)$$

However, cascading comes at a cost: each stage adds series thermal resistance, reducing the absolute forward heat flow:

$$Q_{\text{fwd}}^{(N)} = \frac{\Delta T}{\sum_{i=1}^N R_{\text{fwd},i}} = \frac{\Delta T}{N \cdot R_{\text{fwd},1}} \quad (06.22)$$

The forward conductance decreases linearly with N while rectification grows geometrically, establishing an optimal number of stages for a given application.

Steady-State Limitation

Cascading does not create energy from nothing. A thermal diode does not violate the second law of thermodynamics. At steady state, the temperature distribution adjusts so that the net heat flow is from hot to cold. The diode merely makes the thermal resistance direction-dependent, analogous to an electrical diode making electrical resistance direction-dependent. The rectification ratio quantifies this asymmetry but does not imply perpetual motion or spontaneous cooling.

06.4.3 Performance Benchmarks

Table 06.9 summarizes the current state of the art for thermal diode technologies.

Table 06.9: Thermal diode performance benchmarks (as of 2024). All $\mathcal{R}$ values measured at the stated ΔT ; performance varies strongly with operating conditions.

Technology	$\mathcal{R}$	$\frac{G_{\text{fwd}}}{A}$ (kW/m ² K)	ΔT (K)	TRL	Limitation
Shape-dep. (bulk)	1.1–1.5	10–1000	20–80	5–6	Low $\mathcal{R}$; geometry
VO ₂ thin film	1.5–2.0	1–100	30–50	3–4	Small $\Delta\kappa$; fatigue
GST bilayer	3–5	1–50	50–100	2–3	Slow switch; drift
Near-field (VO ₂)	5–50	0.1–10	10–50	1–2	Gap; area < 1 mm ²
Active (Peltier)	> 100	10–500	any	7–8	Requires power

Benchmark interpretation. The Technology Readiness Level (TRL) follows the NASA SP-2007-6105 scale: TRL 1–3 is laboratory proof-of-concept, TRL 4–6 is component validation in relevant environments, TRL 7–9 is system-level qualification. The rectification ratio $\mathcal{R}$ depends strongly on ΔT : for PCM-based diodes, $\mathcal{R} \rightarrow 1$ when ΔT is too small to trigger the phase transition, and saturates at a material-limited maximum for large ΔT . Forward conductance G_{fwd}/A is normalized per unit area to enable comparison across geometries. Boundary/contact resistances (typically 10^{-6} – 10^{-5} m²K/W at solid-PCM interfaces) are often not separated from bulk G_{fwd} in the literature, which can inflate apparent rectification for thin-film devices. Readers should verify whether published $\mathcal{R}$ values include or exclude interface effects.

Failure Mode:

Thermal Diode Degradation Phase-change thermal diodes suffer from two aging mechanisms: (a) compositional drift during repeated cycling, which shifts T_{switch} and reduces $\Delta\kappa$; and (b) delamination at the PCM-substrate interface due to the volume change during phase transition (~ 5 – 8% for GST). Accelerated testing (1000 thermal cycles, $\Delta T = 100$ K) should demonstrate $< 10\%$ degradation in $\mathcal{R}$ before production deployment.

06.5 Network Context: Switchable and Nonlinear Elements in the Mode-Path Matrix

The Mode-Path Matrix of Chapter 4 assumes constant, linear conductances. Switchable and asymmetric devices break both assumptions: doublets introduce state-dependent conductances, and diodes introduce direction-dependent (nonlinear) conductances. This section extends the MPM formalism to accommodate these devices.

06.5.1 Switchable Devices as Network Reconfiguration

A doublet metadvice on edge E_k of the thermal network graph $\mathcal{G}(V, E)$ replaces the constant conductance G_k with a *tunable* conductance $G_k(\theta)$ given by Eq. (06.6). The network conductance matrix becomes parameter-dependent:

$$\mathbf{G}(\theta) \cdot \mathbf{T} = \mathbf{Q} \quad (06.23)$$

This means the temperature distribution, critical path, and even the Mode-Path Matrix itself depend on the doublet state. The System Design Workflow (Chapter 4, §04.6) must be extended:

1. For each operating state θ_j of the doublet, compute $G_k(\theta_j)$.
2. Rebuild the conductance matrix $\mathbf{G}(\theta_j)$.
3. Re-identify the critical path for each state.
4. Construct a *state-dependent* Mode-Path Matrix: one column per operating state.
5. Verify that the critical path remains conduction-dominated ($\Gamma < 0.1$) in all states—if not, the doublet may be ineffective in some configurations.

Algorithm: MPM Recomputation for State-Dependent Devices

1. **Set state:** For each configuration j , assign θ_j (doublets) or T -bias (diodes).
2. **Update edges:** Recompute $G_k(\theta_j)$ or $G_k^{\text{fwd/rev}}(T)$ for all device-bearing edges (Eq. 06.6 for doublets, Eq. 06.24 for diodes).
3. **Rebuild $\mathbf{G}$:** Assemble updated conductance matrix $\mathbf{G}(\theta_j)$.
4. **Recompute Γ :** For each edge, evaluate $\Gamma_k = G_{\text{rad},k}/G_{\text{cond},k}$ at operating temperature.
5. **Re-identify critical path:** Find the source-to-sink path with maximum total thermal resistance. The critical path may shift between states.
6. **Verify tool selection:** Confirm device remains on critical path *and* Γ supports the device mode (DR-06.4). If not, flag for redesign.

Repeat for each operating state. Store results as columns in the state-dependent MPM table (e.g., Table 06.10).

Conduction Mode (Diffusionics)

Topology reconfiguration. When a doublet switches from concentrator mode ($G_k \gg G_{\text{neighbors}}$) to cloak mode ($G_k \ll G_{\text{neighbors}}$), it can *change the network topology* from the perspective of the critical path. A path that was non-critical may become critical when the doublet blocks it. This is the dynamic analog of DR-T.6 (critical path shift after improvement): here the shift is intentional and controllable.

06.5.2 Thermal Diodes as Nonlinear Network Elements

A thermal diode on edge E_k replaces the constant conductance G_k with a direction-dependent conductance:

$$G_k = \begin{cases} G_{\text{fwd}} & \text{if } T_i > T_j \\ G_{\text{rev}} & \text{if } T_i < T_j \end{cases} \quad (06.24)$$

where nodes i and j are the endpoints of edge E_k , and the “forward” direction is defined as $i \rightarrow j$.

The nonlinear nature of Eq. (06.24) means the standard linear solve $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$ no longer applies directly. Two approaches:

1. **Fixed-point iteration:** Guess $\mathbf{T}$, assign diode states, solve the linear system, check for consistency, and iterate until converged.
2. **Worst-case analysis:** Compute separate MPMs for forward-bias and reverse-bias states, and verify device placement for both.

The placement of a diode in the network determines its system-level impact:

1. **Critical path, forward direction:** The diode is on the critical path and its forward direction aligns with the dominant heat flow. The diode’s high G_{fwd} does not impede normal operation, while G_{rev} blocks parasitic reverse flow. This is the high-impact placement.
2. **Parallel bypass:** The diode sits on a non-critical parallel path. Even perfect rectification ($\mathcal{R} \rightarrow \infty$) on this path has negligible system impact because the parallel path carries only a small fraction of total heat flow.

This reinforces the central lesson of Chapter 4: *where* a device sits in the network matters as much as *what* the device does. The Mode-Path Matrix must be consulted before selecting the diode placement.

Design Rule:

DR-06.4 — Go/No-Go Rule for Switchable and Directional Devices Before deploying a doublet, LMTI, or thermal diode on any network edge E_k , verify *both* conditions:

1. **Critical-path condition:** E_k must lie on the critical path (or within one edge of it). If E_k is on a non-critical parallel bypass, even perfect device performance yields negligible system benefit.
2. **Mode-matching condition:**
 - For *conduction-mode* devices (doublets, LMTI, bulk PCM diodes): require $\Gamma_k < 0.1$ on the target edge. If $\Gamma_k > 0.1$, the radiation channel bypasses the conduction modification and the device benefit degrades as $\sim 1/(1 + \Gamma_k)$.
 - For *radiation-mode* devices (near-field diodes): require $\Gamma_k > 1$ on the target edge. If $\Gamma_k < 1$, the conduction channel dominates and the radiative rectification is irrelevant.

If either condition fails, do not proceed. Redesign the network placement or select a different device class. This rule prevents “device fetishism”—the temptation to deploy an elegant device on a path where it cannot provide system-level benefit.

06.6 Worked Examples

06.6.1 Worked Example WE-06.1: Switchable Wafer Chuck with Doublet Array—Complete MPM Derivation

Problem: A 300 mm wafer chuck processes two alternating recipes: Recipe A requires uniform heating ($T = 350^\circ\text{C} \pm 0.5^\circ\text{C}$), while Recipe B requires a center-to-edge thermal gradient ($\Delta T = 5\text{ K}$, center hot). The chuck currently uses a static resistive heater array embedded in an AlN ceramic body, with He backside gas for wafer-to-chuck coupling. Recipe changes require 30 minutes of thermal stabilization, reducing throughput.

Design a doublet meta-element array integrated into the chuck body that enables switching between Recipes A and B in < 5 minutes, and construct the Mode-Path Matrix to validate the approach.

Solution — The Six-Step System Design Workflow:

Step 1: Define requirements.

Parameter	Recipe A	Recipe B
Temperature	$350 \pm 0.5^\circ\text{C}$	$350 \pm 2.5\text{ K gradient}$
Uniformity	$\pm 0.5^\circ\text{C}$	$\Delta T_{c-e} = 5\text{ K}$
Switching time	$< 5\text{ min}$	
Operating mode	Transparent (uniform)	Concentrator (gradient)

Step 2: Construct thermal network graph.

The five-node network: N1 (plasma), N2 (wafer surface), N3 (wafer bulk), N4 (chuck body), N5 (coolant). Seven edges: E1 (plasma→wafer, radiation), E2 (plasma→wafer, convection), E3 (wafer surface → bulk, conduction), E4 (wafer→chuck, He gas), E5 (chuck heater, internal), E6 (chuck→coolant, conduction through doublet), E7 (parasitic radiation to walls).

Step 3: Compute conductances.

(Representative values per zone, center path.)

Table 06.10: WE-06.1: Edge conductances for the wafer chuck network.

Edge	Path	G_{cond} (W/K)	G_{rad} (W/K)	Γ
E1	Plasma $\rightarrow$ wafer (rad)	—	8	$\gg 1$
E2	Plasma $\rightarrow$ wafer (conv)	—	—	—
E3	Wafer surface $\rightarrow$ bulk	800	0.001	10^{-6}
E4	Wafer $\rightarrow$ chuck (He gas)	80	0.05	6×10^{-4}
E5	Chuck heater (internal)	1000	—	$\ll 0.1$
E6	Chuck $\rightarrow$ coolant (doublet)	500/20	0.02	$< 10^{-4}$
E7	Parasitic (walls)	—	0.3	$\gg 1$

Step 4: Classify edges and identify critical path.

The MPM reveals: E1, E2, E7 are radiation-dominated; E3–E6 are conduction-dominated. The critical path runs $N1 \rightarrow N2 \rightarrow N3 \rightarrow N4 \rightarrow N5$ with bottleneck at E1 (plasma coupling).

Figure 06.7: WE-06.1 Wafer Chuck Network and Mode-Path Matrix

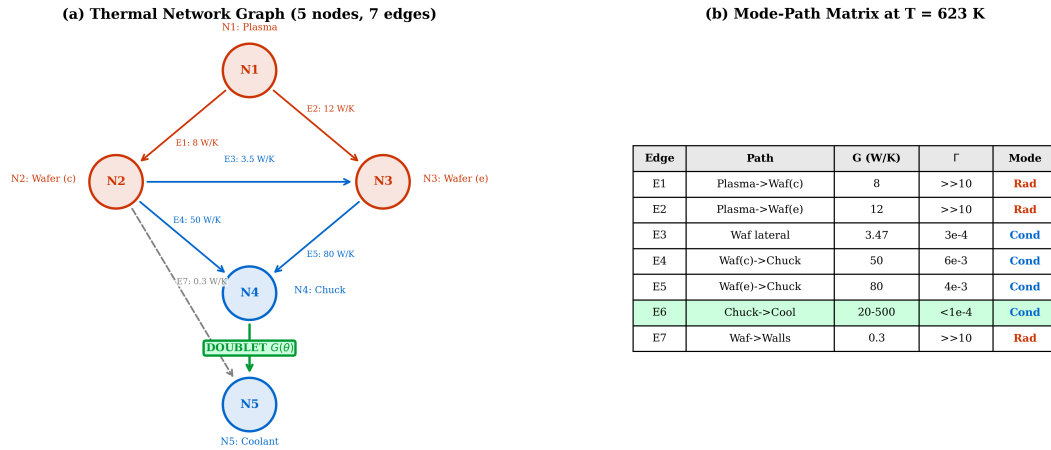


Figure 06.8: WE-06.1: Wafer chuck thermal network and Mode-Path Matrix. (a) Five-node, seven-edge network graph. The doublet array sits on edge E6 (chuck-to-coolant), highlighted in green. Radiation-dominated edges (E1, E2, E7) are shown in red; conduction-dominated edges (E3–E6) in blue. (b) Complete Mode-Path Matrix with Γ classification. Edge E6 is firmly conduction-dominated ($\Gamma < 10^{-4}$), confirming the doublet is the correct tool (DR-05.6 satisfied).

Edge E7 (parasitic radiation to chamber walls) is small ($G_7 = 0.3$ W/K) compared to the main paths and can be neglected in first-order analysis.

Step 5: Apply mode-appropriate tools.

The doublet array sits on E6, which is conduction-dominated. From Eq. (06.7), the required tuning range is $G_{\text{max}}/G_{\text{min}} = 500/20 = 25$.

Fill fraction choice. The doublet's tuning range equals the EMT anisotropy $\kappa_{\parallel}/\kappa_{\perp}$ at the chosen fill fraction (Anisotropy Definitions box, category C). For a transformation thermotics device (cloak or concentrator matching a specific Jacobian), one would use f_{opt} from Eq. (05.15). However, the doublet's purpose here is *maximum tuning range*, not matching a prescribed tensor. We therefore use $f = 0.5$, which maximizes $\kappa_{\parallel}/\kappa_{\perp}$ (Eq. 06.8).

For Si/SiO₂ at $f = 0.5$:

$$\begin{aligned}\kappa_{\parallel} &= 0.5 \times 130 + 0.5 \times 1.4 = 65.7 \text{ W/m}\cdot\text{K} \\ \kappa_{\perp} &= \frac{1}{0.5/130 + 0.5/1.4} = \frac{1}{0.00385 + 0.357} = 2.77 \text{ W/m}\cdot\text{K} \\ \frac{\kappa_{\parallel}}{\kappa_{\perp}} &= \frac{65.7}{2.77} = 23.7 \approx 24\end{aligned}$$

This provides a $24\times$ tuning range, meeting the $25\times$ specification within 4%. The small shortfall is within typical design margin; if the specification is strict, one may either (a) adjust the fill fraction slightly toward the SiO₂-rich side ($f = 0.48$ gives $\kappa_{\parallel}/\kappa_{\perp} \approx 23.6$, so the maximum is broad and insensitive to f near 0.5), or (b) substitute Cu/polyimide ($\kappa_A = 390$, $\kappa_B = 0.35$ W/m·K; $(\kappa_{\parallel}/\kappa_{\perp})_{\max} \approx 280$ at $f = 0.5$), which exceeds the requirement with large margin.

Doublet specification:

- Material pair: Si/SiO₂ ($\kappa_A = 130$, $\kappa_B = 1.4$ W/m·K) at $f = 0.5$
- EMT anisotropy: $\kappa_{\parallel}/\kappa_{\perp} = 24$ (category B' in Anisotropy Definitions box)
- Shell geometry: $R_1 = 3$ mm, $R_2 = 8$ mm (per meta-element)
- Array: 6×6 doublet elements at 25 mm pitch across 150 mm radius
- Interface: Clearance + Galinstan LMTI (DR-06.1, $d = 30$ μm gap)
- G_{LMTI} per element: $16.5 \times 5 \times 10^{-4} / (30 \times 10^{-6}) = 275$ W/K $\gg G_6(\pi) = 20$ W/K. Interface is not the bottleneck (DR-06.2 satisfied).

Recipe A (uniform): Set $\theta = \pi/2$ (transparent mode). All doublets present equal conductance: $G_{\text{eff}} \approx (\kappa_{\parallel} + \kappa_{\perp})/2 \cdot \text{geometry} \approx 160$ W/K per element.

Recipe B (gradient): Set center elements to $\theta = 0$ (concentrator, $G_{\text{eff}} = 500$ W/K) and edge elements to $\theta = \pi$ (cloak, $G_{\text{eff}} = 500/24 \approx 21$ W/K). The $24\times$ conductance ratio creates the required center-to-edge gradient.

Conduction Mode (Diffusionics)

Why $f = 0.5$ and not f_{opt} for the doublet. The transformation-device fill fraction $f_{\text{opt}} = 1/(1 + \sqrt{\kappa_A/\kappa_B})$ is designed to match a *specific* pair of $\kappa_{\parallel}$ and $\kappa_{\perp}$ values prescribed by the coordinate-transformation Jacobian (Chapter 5, Eqs. 05.3–05.5). The doublet, however, does not need to match any prescribed tensor; it needs the *widest possible tuning range*. Since the tuning range $G_{\max}/G_{\min} = \kappa_{\parallel}/\kappa_{\perp}$, and $\kappa_{\parallel}/\kappa_{\perp}$ is maximized at $f = 0.5$ (Eq. 06.8), doublet designers should always use $f = 0.5$ unless constrained by other factors (IR transparency, mechanical thickness, etc.). For Si/SiO₂, the difference is dramatic: $f = 0.5$ gives $24\times$ tuning range, while $f_{\text{opt}} = 0.094$ gives only $8.7\times$.

Step 6: Verify and iterate.

After setting Recipe B doublet states, recalculate the network:

Center path: $R_A = 0.125 + 0.020 + 0.002 = 0.147$ K/W (bottleneck: E1, plasma coupling).

Edge path: $R_B = 0.083 + 0.0125 + 0.050 = 0.146$ K/W (bottleneck shifts to E6 at edge, the cloaked doublet).

The two paths are now nearly balanced ($R_A \approx R_B$), with the center running hotter by $\Delta T \approx Q \cdot (R_A - R_B) \approx 0.1$ K difference from path resistance alone—but the doublet conductance asymmetry ($G_{6,\text{center}} = 500$ vs. $G_{6,\text{edge}} = 20$ W/K) creates the desired 5 K gradient by controlling how efficiently each zone dumps heat to the coolant.

Switching time estimate: The thermal time constant depends on the thermal mass under consideration. For a *single meta-element zone* (25 mm pitch tile, AlN body: $\rho = 3260 \text{ kg/m}^3$, $c_p = 740 \text{ J/(kg}\cdot\text{K)}$, volume $\approx 25 \times 25 \times 10 \text{ mm}^3 = 6.25 \times 10^{-6} \text{ m}^3$):

$$C_{\text{zone}} = \rho c_p V \approx 3260 \times 740 \times 6.25 \times 10^{-6} \approx 15 \text{ J/K}$$

With $G_6 \approx 160 \text{ W/K}$ per element (transparent mode), the local time constant is $\tau_{\text{zone}} = C_{\text{zone}}/G_6 \approx 0.09 \text{ s}$. Reaching 3τ (95% settling) takes $\approx 0.3 \text{ s}$ per zone—fast enough for individual element switching.

For the *full chuck body* (300 mm diameter, 25 mm thick AlN, $V \approx 1.77 \times 10^{-3} \text{ m}^3$):

$$C_{\text{chuck}} = 3260 \times 740 \times 1.77 \times 10^{-3} \approx 4270 \text{ J/K}$$

With aggregate coolant conductance $G_{\text{cool}} \approx 300 \text{ W/K}$, $\tau_{\text{chuck}} \approx 14 \text{ s}$, and $3\tau_{\text{chuck}} \approx 43 \text{ s}$ for full equilibration. The 5-minute switching specification is met with comfortable margin.

Steady-State Limitation

Local vs. global time constants. The doublet array enables zone-by-zone switching (sub-second per element) even though the full chuck body equilibrates in $\sim 14 \text{ s}$. The wafer's own thermal time constant ($\tau_{\text{wafer}} \approx 0.05\text{--}0.2 \text{ s}$ for a $775 \text{ }\mu\text{m}$ Si wafer) is faster than both. The system bottleneck is therefore the chuck thermal mass, not the metadvice actuation.

Lesson: The MPM reveals that the doublet is correctly placed on a conduction-dominated path (E6) but cannot address the radiation-dominated paths (E1, E2).

06.6.2 Worked Example WE-06.2: Cascaded GST Thermal Diode for 3D IC Package

Problem: A 3D-stacked IC package with two high-power dies ($P_1 = 5 \text{ W}$, $P_2 = 3 \text{ W}$) connected by solder bumps ($G_{\text{bump}} = 10 \text{ W/K}$) requires thermal rectification: heat must flow from the die to the heat sink, but reverse flow from the board during power-off transients must be suppressed by at least $20\times$.

Solution: Design a cascaded GST thermal diode with $\mathcal{R} > 20$ and $G_{\text{fwd}} > 2 \text{ W/K}$.

Each GST stage: $\mathcal{R}_1 = 3.5$, $G_{\text{fwd},1} = 30 \text{ W/K}$

($5 \text{ }\mu\text{m}$ GST, area = 1 cm^2).

Number of stages: $\mathcal{R}^{(N)} = 3.5^N > 20 \Rightarrow N \geq \log(20)/\log(3.5) = 2.4$. Use $N = 3$ stages: $\mathcal{R}^{(3)} = 3.5^3 \approx 43$. Pass.

Forward conductance: $G_{\text{fwd}}^{(3)} = 30/3 = 10 \text{ W/K} > 2 \text{ W/K}$. Pass.

MPM check: The die-to-substrate path operates at $T \approx 85^\circ\text{C}$ (358 K). $G_{\text{cond}} = 10 \text{ W/K}$, $G_{\text{rad}} = 4 \times 0.3 \times 5.67 \times 10^{-8} \times 358^3 \times 10^{-4} = 3.1 \times 10^{-4} \text{ W/K}$. $\Gamma = 3.1 \times 10^{-5} \ll 0.1$. Firmly conduction-dominated. The GST diode (a conduction-mode device) is appropriate.

Integration: The three GST stages are deposited as a thin-film trilayer ($3 \times 5 \text{ }\mu\text{m} = 15 \text{ }\mu\text{m}$ total) between the die backside metallization and the solder bump pads, with SiN_x diffusion barriers (50 nm each) between stages. Total added thickness: $15.3 \text{ }\mu\text{m}$ —compatible with standard packaging processes.

Figure 06.8: WE-06.2 Cascaded GST Thermal Diode

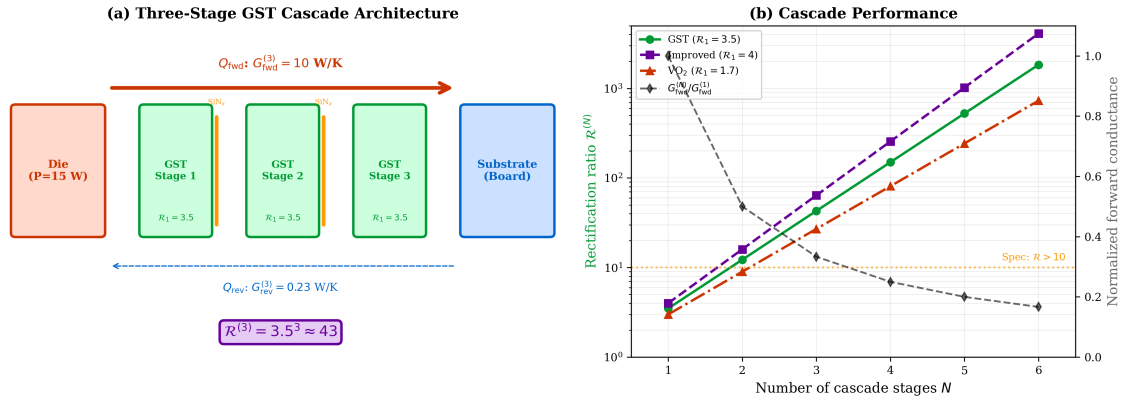


Figure 06.9: WE-06.2: Cascaded GST thermal diode. **(a)** Three-stage architecture: each GST layer ($\mathcal{R}_1 = 3.5$) is separated by SiN_x diffusion barriers. Forward flow (Q_{fwd} , $G_{\text{fwd}}^{(3)} = 10 \text{ W/K}$) is large; reverse flow (Q_{rev} , $G_{\text{rev}}^{(3)} = 0.23 \text{ W/K}$) is suppressed by $\mathcal{R}^{(3)} \approx 43$. **(b)** Performance scaling: $\mathcal{R}^{(N)}$ grows geometrically (left axis) while $G_{\text{fwd}}^{(N)}$ decreases linearly (right axis), establishing a trade-off with an optimal N depending on the application specification.

Chapter Summary

This chapter introduced two families of dynamic conduction devices that extend the static toolkit of Chapter 5:

1. **Doublet metadevices** (§06.1): Counter-rotating anisotropic shells that continuously tune between cloaking, transparent, and concentrating modes. The tuning range equals the constituent anisotropy ratio $\kappa_{\parallel}/\kappa_{\perp}$ (Eq. 06.7). Production-grade implementation requires LMTI rotating joints (§06.2, DR-06.1, DR-06.2).
2. **Thermal diodes** (§06.3): Devices that rectify heat flow, with mechanisms ranging from shape-dependent ($\mathcal{R} \approx 1.5$) to phase-change materials ($\mathcal{R} \approx 3\text{--}5$) to near-field radiative ($\mathcal{R} > 10$). Cascading N stages multiplies rectification geometrically (Eq. 06.20).
3. **Network integration** (§06.5): Switchable devices create state-dependent conductance matrices $\mathbf{G}(\theta)$; diodes create direction-dependent (nonlinear) matrices $\mathbf{G}(\mathbf{T})$. Both require the MPM to be recomputed for each operating state.
4. **MPM validation** (WE-06.1): The Mode-Path Matrix confirms that doublets and diodes are effective only on conduction-dominated paths ($\Gamma < 0.1$). Placing them on radiation-dominated paths wastes resources.

Production Readiness Checklists

PRC-1: Doublet Metadvice

Requirement	Specification
Fill fraction	$f = 0.5$ for max tuning range (cat. B'); f_{opt} for Jacobian matching (cat. B)
Tuning range	$\kappa_{\parallel}/\kappa_{\perp} \geq G_{\text{max}}/G_{\text{min}}$ required
Interface type	Clearance + LMTI for > 10 cycles (DR-06.1)
LMTI preparation	$N_i \geq 2 \mu\text{m}$, $R_a < 0.8 \mu\text{m}$, inert assembly (DR-06.2)
Interface overhead	$G_{\text{LMTI}} > 10 \times G_{\text{doublet,min}}$
Cycling endurance	$\geq 10^4$ cycles with $\Delta G/G < 15\%$
Array layout	Min 4×4 ; pitch $\leq$ spatial target (DR-06.3)
MPM validation	Target edge: $\Gamma < 0.1$ and on critical path (DR-06.4)

PRC-2: PCM Thermal Diode (VO₂/GST)

Requirement	Specification
Rectification	$\mathcal{R} > \mathcal{R}_{\min}$; cascade N stages if needed (Eq. 06.20)
Forward conductance	$G_{\text{fwd}}^{(N)} > G_{\text{fwd},\min}$
Transition temperature	T_{switch} within operating ΔT window
Diffusion barriers	SiN _x or Al ₂ O ₃ between stages (≥ 50 nm)
Cycling endurance	$\geq 10^3$ cycles; $\Delta\mathcal{R}/\mathcal{R} < 10\%$, $\Delta G_{\text{fwd}}/G_{\text{fwd}} < 10\%$
Delamination check	Volume change (5–8% for GST) verified by acoustic microscopy
MPM validation	$\Gamma < 0.1$ on critical path (DR-06.4)

PRC-3: Near-Field Radiative Diode (Research Only)

Requirement	Specification
Gap distance	50–500 nm; active control needed
Planarity	± 10 nm; CMP-grade surfaces
Max area	$< 1 \text{ mm}^2$ (current limit)
Contamination	Cleanroom Class 100 assembly
Γ validation	$\Gamma > 1$ on target edge; no conduction shunting
TRL status	1–2. Lab only.

Table 06.11: Complete set of Chapter 6 design rules.

Rule	Section	Description
DR-06.1	§06.1.4	Doublet interface selection: use clearance + LMTI for > 10 switching cycles; verify $\Delta R_{\text{th}}/R_{\text{th}} < 0.15$.
DR-06.2	§06.2.3	LMTI surface preparation: Ni barrier $\geq 2 \mu\text{m}$; $R_a < 0.8 \mu\text{m}$; inert atmosphere assembly; verify $G_{\text{LMTI}} > 10 \times G_{\text{doublet},\min}$.
DR-06.3	§06.4.1	Planar doublet array design: min 4×4 array; pitch $\leq$ target spatial resolution; include boundary compensation.
DR-06.4	§06.5	Go/no-go rule for switchable devices: deploy only on conduction-dominated critical-path edges ($\Gamma < 0.1$); otherwise benefit is marginal.

Looking Forward

Chapter 7 translates the cylindrical and planar devices of Chapters 5 and 6 into production-ready architectures using via arrays—the “production thermal metamaterial.” Through-silicon vias (TSVs) and through-glass vias (TGVs) achieve anisotropy ratios of 10:1 to 100:1 using standard semiconductor fabrication processes, bridging the gap between laboratory metamaterial demonstrations and fab-ready hardware.

The doublet arrays of DR-06.3 find their practical realization in Chapter 11 (Semiconductor Thermal Control), where the smart wafer chuck concept integrates doublet meta-elements with closed-loop temperature control for $\pm 0.1^\circ\text{C}$ uniformity.

The thermal diodes of Section 06.3 reappear in Chapter 14 (The Quantum-Thermal Frontier), where the diode’s directional asymmetry is *inverted*: instead of enhancing forward dissipation, the quantum application requires that heat can *leave* the qubit but cannot *enter*—the same physics, opposite optimization target (Table 14.15).

Research Roadmap

Table 06.12: Research roadmap for Chapter 6 topics.

Topic	Current State	5-Year Horizon	TPD Connection
Doublet cycling	Lab demos, < 1000 cycles	Production-grade $> 10^5$ cycles with LMTI	Ch. 11 (wafer chuck)
GST thermal diodes	$\mathcal{R} \approx 3\text{--}5$, slow switching	$\mathcal{R} > 10$, < 1 ms switch	Ch. 12 (failure analysis)
Near-field diodes	Gap ~ 100 nm, $< 1\text{ mm}^2$ area	Wafer-scale gap maintenance	Ch. 3, Ch. 13
Electrically switchable PCM diodes	Joule-heated GST demos	Integrated PCM-CMOS thermal switches	Ch. 7, Ch. 15
ML-optimized doublet arrays	Manual angle setting	Real-time adaptive control	Ch. 15

Problems

1. **Problem 6.1: Doublet Tuning Curve.** A doublet metadvice uses Cu/Kapton shells ($\kappa_A = 390$ W/m·K, $\kappa_B = 0.12$ W/m·K, $R_1 = 5$ mm, $R_2 = 15$ mm, $L = 20$ mm). (a) Compute $G_{\text{eff}}(\theta)$ at $\theta = 0, \pi/6, \pi/3, \pi/2, 2\pi/3, 5\pi/6$, and π using Eq. (06.6). (b) Plot $G_{\text{eff}}(\theta)/G_{\text{eff}}(0)$ and verify the $\cos^2(\theta/2)$ dependence. (c) What is the tuning range $G_{\text{max}}/G_{\text{min}}$? (d) If a 50 μm Galinstan LMTI gap is present at R_1 , compute G_{LMTI} and determine whether the interface limits the tuning range.
2. **Problem 6.2: LMTI Degradation.** A doublet with Galinstan LMTI ($d = 30$ μm , $A = 8 \times 10^{-4}$ m²) has Ni barrier thickness $t_{\text{Ni}} = 2$ μm . After 6 months at 200°C , the Ni barrier partially fails and a Ga-Al intermetallic layer ($\kappa = 30$ W/m·K) of thickness 5 μm forms. (a) Compute the initial G_{LMTI} . (b) Compute the degraded G_{LMTI} with the intermetallic layer in series. (c) What percentage of the original conductance is lost? (d) At what intermetallic thickness does G_{LMTI} drop below $10 \times G_{\text{doublet,min}}$ (DR-06.2 violation), assuming $G_{\text{doublet,min}} = 5$ W/K?
3. **Problem 6.3: Rectification Ratio.** A VO₂-based thermal diode bilayer consists of 100 μm VO₂ in series with 100 μm SiO₂ (area = 1 cm²). The hot side is at $T_H = 90^\circ\text{C}$ and the cold side at $T_C = 50^\circ\text{C}$. (a) Compute G_{fwd} (VO₂ metallic, $\kappa = 6$ W/m·K; SiO₂, $\kappa = 1.4$ W/m·K). (b) Compute G_{rev} (VO₂ insulating, $\kappa = 3.5$ W/m·K). (c) Compute $\mathcal{R}$. (d) What happens if ΔT is reduced so that neither side crosses the MIT? What is $\mathcal{R}$ then?
4. **Problem 6.4: Cascade Optimization.** A packaging application requires $\mathcal{R} > 20$ with $G_{\text{fwd}} > 2$ W/K. Each GST diode stage has $\mathcal{R}_1 = 4$ and $G_{\text{fwd},1} = 15$ W/K. (a) Compute the minimum number of stages N . (b) Compute $G_{\text{fwd}}^{(N)}$ and verify it meets the conductance specification. (c) If $G_{\text{fwd}}^{(N)}$ is insufficient, propose a design modification (hint: increase area or reduce thickness) and recalculate. (d) Plot $\mathcal{R}^{(N)}$ and $G_{\text{fwd}}^{(N)}$ vs. N for $N = 1$ – 6 . Identify the optimal N .
5. **Problem 6.5: MPM with Thermal Diode.** A 3-node series network: Die (N1, $Q = 10$ W) $\rightarrow$ TIM (N2) $\rightarrow$ Heat Sink (N3, $T_3 = 25^\circ\text{C}$). $G_{12} = 20$ W/K (standard TIM), $G_{23} = 50$ W/K. A thermal diode is inserted at G_{12} : $G_{\text{fwd}} = 20$ W/K, $G_{\text{rev}} = 2$ W/K. (a) Compute T_1 and T_2 in normal operation (forward bias). (b) If the heat sink temporarily heats to $T_3 = 80^\circ\text{C}$ while $Q = 0$ (power-off transient), compute T_1 with and without the diode. (c) Compute Γ for the G_{12} path at $T = 350$ K. Is the diode on a conduction-dominated path? (d) If $\Gamma_{12} = 0.5$, would a conduction-mode diode still be effective? Explain.
6. **Problem 6.6: Doublet State-Dependent MPM.** Using the wafer chuck network from WE-06.1, construct the complete Mode-Path Matrix for three doublet states: $\theta = 0$ (concentrator), $\theta = \pi/2$ (transparent), and $\theta = \pi$ (cloak). For each state: (a) Compute all seven edge conductances. (b) Identify the critical path and bottleneck element. (c) Compute Γ for each edge. (d) Present the results as three columns in a single MPM table. (e) Does the critical path shift between states? Does the mode classification of any edge change?
7. **Problem 6.7: Near-Field Diode MPM Assessment.** A proposed near-field VO₂ diode for a 3D IC package has: gap $d = 200$ nm, area $A = 1$ mm², $\mathcal{R} = 15$ (measured). The diode sits between Die 1 (bottom, $P = 5$ W) and Die 2 (top, $P = 3$ W) in a face-to-face bonded stack. (a) Estimate G_{rad} across the 200 nm gap at $T = 350$ K using the near-field enhancement factor $\sim 100\times$ over blackbody. (b) Estimate G_{cond} through the surrounding bonding material ($\kappa = 200$ W/m·K, $A = 99$ mm², $L = 200$ nm). (c) Compute

- Γ for the diode path and for the bonding-material path. Is the near-field diode shunted?
- (d) Using DR-06.4, determine whether the near-field diode provides system-level benefit.

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Chapter 6 of “Thermal Photonics: A Unified Framework for Engineering Heat Flux”

Book 3 of the Open-Source Engineering Optics Trilogy

Cross-references: Eikonal Bridge Ch. 8 (optical isolator analogy), Ch. 9 (zoom lens tolerancing); QFI

Ch. 6 (detector thermal management)